

MEDIATEK

MT8676 Android Thermal User Manual

Version: 1.0
Release date: 2024-08-12

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Version History

Version	Date	Author	Description
1.0	2024-08-12	Zhaoqing Jiu	Official release

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1 Thermal

1.1 Overview

Thermal management on device has two main goals:

- Controls component temperature to avoid being damaged by heat
- Controls the temperature of the whole product for human safety and meet safety regulation

Heat in the device is accumulated from high power of ICs. High power is side effect of high clock speed, voltage, and performance.

The approach to control temperature from thermal management is to control power dissipation and consumption. Power dissipation control can be achieved by adding various thermal solutions such as TIM, copper foil, tube, etc to efficiently spread the heat to the whole product and to the air or contact surface. Power consumption control can be achieved by reducing/limiting power consumption based on temperature which is the focus of this document.

1.2 Architecture/Process Overview

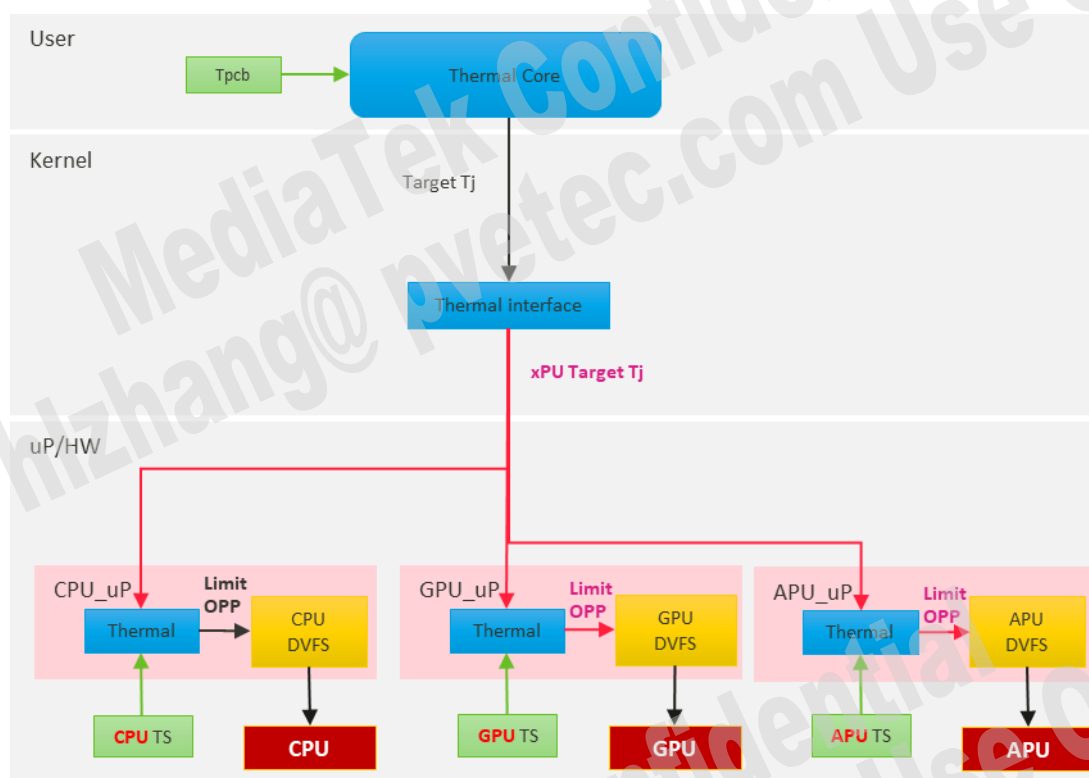


Figure 1-1. Thermal 2.0 software architecture

Thermal_core is a native Linux application. Its main function is to parse a thermal policy of the .conf format.

CPU/GPU/APU will each perform thermal throttling based on the target Tj set by thermal_core and the monitored temperature.

1.2.1.1 Thermal HAL

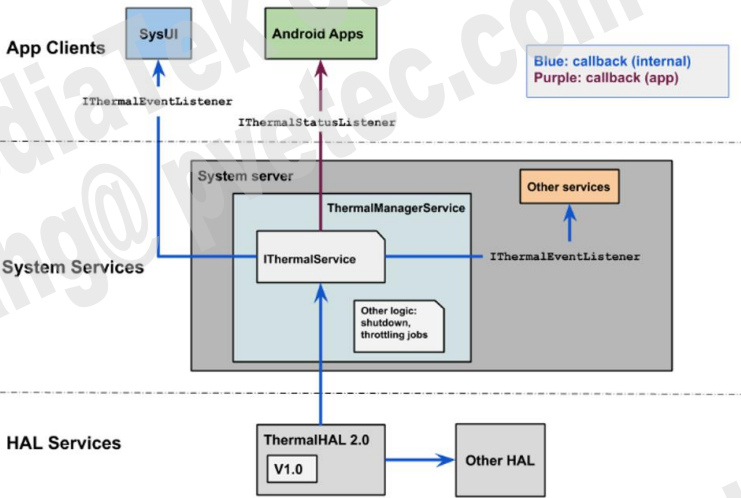


Figure 1-2. Thermal HAL

- A native layer service that offers thermal info API for application layer or other native service
- Get thermal zone temperature via thermal sysfs node
- Some applications or modules need to get thermal information, ex: cpu/gpu/tskin temperatures, cpu usage, cooling device list
- Google defines some HAL interfaces that chip vendors must implement, such as temperature query or thermal status change notification

1.3 Configuration/Customization Guideline

1.3.1 Thermal Policy

- Policy path in codebase
vendor/mediatek/proprietary/external/thermal_core_lib/mt6897/
- Policy Template
vendor/mediatek/proprietary/external/thermal_core_lib/thermal_policy_template.conf
- Supports encrypted format

Table 1-1. Available thermal policies on device (/vendor/etc/thermal/)

Thermal policy	Permanent?	Encrypted?	Description
thermal.conf	Yes	Yes	Default thermal policy
disable_thermal.conf	Yes	Yes	Disable thermal throttling and thermal protection
disable_thermal_temp.conf	No	Yes	Same as the above, except it need to re-apply after device rebooted.
disable_throttling.conf	No	Yes	Disable thermal throttling
disable_skin_control.conf	No	Yes	Disable MTK skin control close loop (always keep Target Tj to 95°C)

Thermal policy	Permanent?	Encrypted?	Description
Thermal_policy_XX.conf (XX = 00~19 except 00, 02, 08)	On demand	Yes	Can add your own policy setting and switch via power HAL
thermal_policy_08.conf	No	Yes	Thermal policy for benchmark

1.3.1.1 Thermal Policy Commands

- Apply a thermal policy

```
adb shell "thermal_intf apply [policy_name]"
```

E.g. adb shell "thermal_intf apply disable_throttling.conf"

- Check the current thermal policy

```
adb shell cat /data/vendor/thermal/.current_tp
```

- Check the current permanent policy

```
adb shell cat /data/vendor/thermal/.permanent_tp
```

1.3.1.2 Thermal Policy Format

- Permanent policy

```
[policy_type]
# If permanent= Yes, this policy is still valid after reboot
permanent= Yes
```

- Linux thermal framework (LTF)

- "policy": Support power_allocator and step_wise.

```
[LTF]
thermal_zone= soc_max
mode= enabled
policy= power_allocator
switch_on_temp= 85000
control_temp= 95000
critical_temp= 115000
cooler= [thermal-cpufreq-0, thermal-devfreq-0]
cooler_weight= [1024, 1024]
```

- Disable LTF throttling

```
[LTF-disable-throttling]
# If HW_protection= disabled, it will turn off
# SW shutdown protections and LVTS HW reboot protection
HW_protection= enabled
```

- Disable LTF shutdown cooler and LVTS thermal reboot

```
[LTF-disable-throttling]
# If HW_protection= disabled, it will turn off
# SW shutdown protections and LVTS HW reboot protection
HW_protection= disabled
```

- Closed loop Tskin control

- trip_pcb: PCB temperature to enable closed loop.
- target_tpcb: Target PCB temperature for closed loop.

```
[mtk-skin-control]
max_target_tj= 95000
min_target_tj= 63000
trip_tpcb= 47000
target_tpcb= 50000
```

- **Backlight cooler**

- “reduce-brightness”: Reduce brightness xx %.

```
[backlight-cooler]
mode= enabled
sensor= ap_ntc
threshold= [43000, 46000, 49000, 53000]
hysteresis= [2000, 2000, 2000, 2000]
reduce-brightness= [10, 50, 80, 95]
```

- **CPU frequency table mapping**

- “cluster”: CPU cluster id.

```
[mtk-cpu-table-mapping]
mode= enabled
sensor= ap_ntc
cluster= 0
threshold= [39000, 42000, 44000, 46000, 48000, 50000, 52000]
hysteresis= [2000, 2000, 2000, 2000, 2000, 2000, 2000]
freq= [2000000, 1800000, 1800000, 1800000, 1800000, 1725000, 1725000]
```

- **CPU core isolation table mapping**

- “CPU”: CPU core to be isolated.

```
[mtk-cpu-isolate]
mode= enabled
sensor= ap_ntc
cpu= [cpu4, cpu5, cpu6, cpu7]
threshold= [52000, 52000, 48000, 48000]
hysteresis= [2000, 2000, 2000, 2000]
```

- **GPU frequency table mapping**

```
[mtk-gpu-table-mapping]
mode= enabled
sensor= ap_ntc
threshold= [39000, 42000, 44000]
hysteresis= [2000, 2000, 2000]
freq= [854000, 815000, 715000]
```

- **Update thermal HAL severity levels**

- Support only SKIN type
- “level”: level can only be severe, critical, emergency, shutdown.

```
[thermalhal-threshold]
type= SKIN
level= [severe, critical, emergency, shutdown]
threshold= [50000, 56000, 58000, 70000]
```

- **Fan Cooler (in dts file)**

- “pwm-ch”: pwm to control fan.


```

&fan_cooler {
    pinctrl-names = "default", "active";
    pinctrl-0 = <&pwm0_gpio_inactive_cfg>;
    pinctrl-1 = <&pwm0_gpio_active_cfg>;
    pwm-ch = <3>; /* GPIO 94 */
    status = "okay";
};

&pio {
    pwm0_gpio_inactive_cfg: pwm0inactive {
        pins-cmd-dat {
            pinmux = <PINMUX_GPIO94_FUNC_PWM_3>;
            input-high;
        };
    };

    pwm0_gpio_active_cfg: pwm0active {
        pins-cmd-dat {
            pinmux = <PINMUX_GPIO94_FUNC_PWM_3>;
            output-high;
        };
    };
};

```

- Bind to thermal_zone.

```

&thermal_zones {
    soc_max {
        polling-delay = <500>; /* milliseconds */
        polling-delay-passive = <500>; /* milliseconds */

        trips {
            soc_warm: soc-warm {
                temperature = <65000>;
                hysteresis = <2000>;
                type = "active";
            };

            soc_mid: soc-mid {
                temperature = <75000>;
                hysteresis = <2000>;
                type = "active";
            };

            soc_hot: soc-hot {
                temperature = <85000>;
                hysteresis = <2000>;
                type = "active";
            };
        };
    };

    cooling-maps {
        map0 {
            trip = <&soc_warm>;
            cooling-device = <&fan_cooler 1 1>;
        };

        map1 {
            trip = <&soc_mid>;
            cooling-device = <&fan_cooler 2 2>;
        };

        map2 {
            trip = <&soc_hot>;
            cooling-device = <&fan_cooler 3 3>;
        };
    };
};

```


1.4 Frequently Asked Questions/Troubleshooting

Turn on more thermal_core log:

```
adb shell "/vendor/bin/thermal_intf debug_log 1"
```

How to decrypt or encrypt thermal configuration files:

<https://online.mediatek.com/apps/faq/detail?list=HW&faqid=FAQ27718>

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